

Modality 3: Gesture

Brief: Gesture Controlled Satellite

Description:

Another task astronauts might have to do on their missions is operate unmanned vehicles to get to places beyond their reach. This week, we'll learn to control the movement of a satellite remotely using different gestural methods.

Resources:

Google MediaPipe:

https://developers.google.com/edge/mediapipe/solutions/vision/hand_landmarker

<https://developers.googleblog.com/drone-control-via-gestures-using-mediapipe-hands/>

Custom Gesture Models:

<https://www.thirteen23.com/updates/speaking-in-gestures>

<https://medium.com/@ohr.morris/leveraging-google-mediapipe-for-easy-hand-gesture-classification-8ff2365c202c>

Arduino:

<https://projecthub.arduino.cc/Isaac100/getting-started-with-the-hc-sr04-ultrasonic-sensor-7cabe1>

https://www.sameskydevices.com/blog/the-basics-of-ultrasonic-sensors?srltid=AfmBOopa6gYkofER97FtEOO_jShVR30iKW22G1eMW84re_eeOF9VKY6B

Processing Documentation:

<https://processing.org/examples/loaddisplayimage.html>

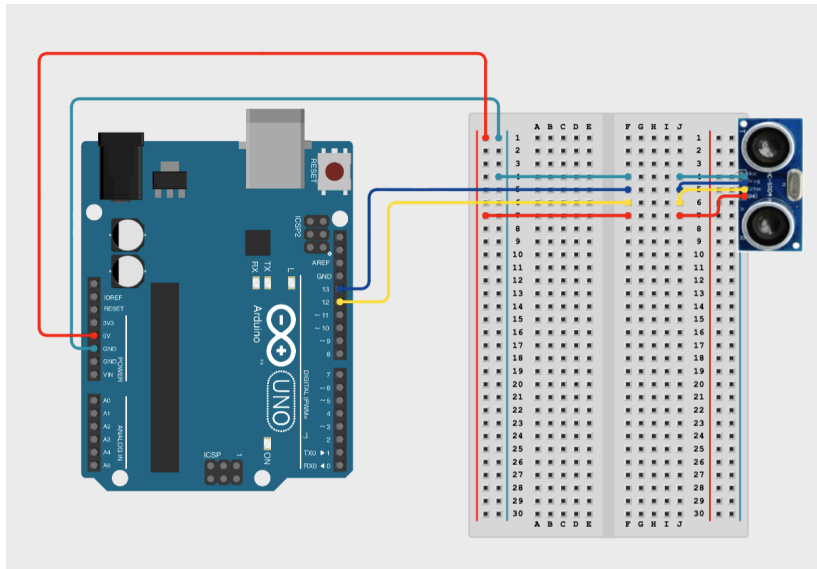
<https://processing.org/examples/words.html>

<https://processing.org/examples/alphamask.html>

Readings:

https://www.sas.upenn.edu/~cavitch/pdf-library/Nagel_Bat.pdf

Example 1: Ultrasonic Sensor



What's Happening:

1. Ultrasonic sensor connected to Arduino detects distance from objects.
2. Distance value is sent to processing via serial port.
3. Distance value is used to control the scale of the satellite overlay in Processing, giving the illusion of the satellite moving closer/further away.

Arduino Connections:

1. With the Arduino unpowered, connect the GND on the power side of the arduino to the ground strip on the breadboard.
2. Connect the 5v to the power strip on the breadboard
3. Connect 5V power to the VDD pin on the ultrasonic sensor
4. Connect GND to the GND pin on the sensor
5. Connect trig pin to Digital Pin [13]
6. Connect echo pin to Digital Pin [12]

Arduino Code:

1. On your computer, open the Arduino IDE
2. Plug the Arduino in via USB
3. Open the example file Gesture_AR.ino
4. Remember to take note of your Arduino port.
5. Click the upload button to send the code to your Arduino.
6. If it is working, you should see the distance value in the serial monitor change depending on how far the closest object is in front of the ultrasonic sensor.

Processing Instructions:

1. Open the Processing software
2. Open the example file GESTURE_E1.pde
3. Change the Arduino port to match the one you noted before.
4. Press the play button
5. If all goes well, you should be able to now move the satellite towards and away from you based on the position of your hand.
6. Some parameters you can play with include: minDistance, maxDistance, easing

Example 2: Gesture-Based Movement



What's Happening:

1. Gesture Recognition via running Google Mediapipe via Python
2. Two gestures are tracked: thumbs up and thumbs down.
3. Thumbs Up moves the satellite up in the Y-Axis, and Thumbs Down does the opposite.
4. Gesture percentages are sent via OSC.
5. OSC data is sent via serial port
6. OSC data is sent to a Processing sketch via the OSCp5 library.
7. Processing sketch has two images set up: a background and an overlay mask.
8. The overlay mask is moved using the coordinates sent via OSC.

Python Instructions:

1. Make sure python and the relevant mediapipe libraries are installed as per the instructions in Canvas.
2. In Terminal, navigate to the folder where your VISION folder is installed by using the commands:

"ls" to list folders in the directory

"cd foldername" to navigate to folders within the directory

"cd .." to return to the parent directory

3. Once you are in the folder, enter:

Python3 or pip3 GESTURE.py

4. If everything is installed correctly, a window should pop up, which labels your hands according to the following gestures:

0 - Unrecognized gesture, label: Unknown

1 - Closed fist, label: Closed_Fist

2 - Open palm, label: Open_Palm

3 - Pointing up, label: Pointing_Up

4 - Thumbs down, label: Thumb_Down

5 - Thumbs up, label: Thumb_Up

6 - Victory, label: Victory

7 - Love, label: ILoveYou

5. To close this window, press Q.

Troubleshooting:

if you are having trouble with Python and mediapipe working, ensure you have right libraries installed, including opencv-python and mediapipe.

- An import cv2 error normally results from not having opencv-python
- Sometimes version errors occur. This works best with Python 3.10.x. You might also have to change to a different version of mediapipe. Version 0.10.x

Processing Instructions:

1. Open the Processing software
2. Open the example file GESTURE_E2.pde
3. Press the play button, with the python code running.
4. If all goes well, you should be able to now move the satellite up and down in the window with Thumbs Up and Thumbs Down Gestures.

TO THINK ABOUT:

1. What are appropriate gestures for different kinds of movements?
2. Who might benefit from gesture recognition in interfaces and what are some drawbacks?
3. Where might you draw inspiration for new kinds of gestures?
4. When might you want to use gesture/CV or a more basic form of recognition like an ultrasonic sensor?

Challenges:

1. Example 2 works with movement in the Y-axis at present. Modify this script to allow for additional movement in the X-axis with a different gesture.
2. Combine the codes from the two examples so that you can move the satellite in 3-axes.
3. Add a custom gesture which allows you to rotate the satellite.
4. Come up with an additional example of how you might use a gesture-based control interface in an everyday setting. Why might this be useful?
5. Think of some gestures in the natural world, how might you incorporate these as interactions?